

There Must Be Some Kind of Mistake: Distinguishing Errors from Uncertainty in Type Checking

This paper extends work of Garby, Bahr, and Hutton on calculating type checkers, to distinguish errors that will certainly occur at runtime from uncertainty due to the approximate nature of type checking. The work also extends this to a simple situation with possibly diverging programs, where again we distinguish certainty of divergence from uncertainty due to approximation.

CCS Concepts: • **Software and its engineering** → **Formal software verification**; • **Theory of computation** → **Program verification**; **Type theory**.

Additional Key Words and Phrases: type checking, nontermination, type errors

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1 INTRODUCTION

Garby et al. [1] presents the derivation of a type checker for a simple expression language, in the style of program calculation. One essentially tries to prove a specification about code that has yet to be determined, and discovers the definitions (or constraints on them) necessary in the course of attempting to complete the proof. Their specification states that the type computed by the type checker texp for an *Expr* e is an approximation to the true type of e . The true type is the type computed (by a function tval) for the value of e , as produced by an evaluator eval . The notion of approximation is that the *ERROR* type returned by the type checker to signal a type error approximates any of the types for values. With the types shown in Figure 1, their specification is

$$\text{texp } e \leq \text{tval } (\text{eval } e)$$

REFERENCES

- [1] Zac Garby, Patrick Bahr, and Graham Hutton. 2025. The Calculated Typer (Functional Pearl). In *Proceedings of the 18th ACM SIGPLAN International Haskell Symposium* (Singapore, Singapore) (*Haskell '25*). Association for Computing Machinery, New York, NY, USA, 17–29. <https://doi.org/10.1145/3759164.3759346>

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data **Type** = **INT** | **BOOL** | **ERROR**

Fig. 1. Definitions from Garby et al. [1]
xO